Question Answer Session

21 Kasım 2020 Cumartesi 12:31

*3. Show algebraically that all the basic solutions of the following LP are infeasible.

$$\text{Maximize } z = x_1 + x_2$$

subject to

$$x_1 + 2x_2 \le 6$$
$$2x_1 + x_2 \le 16 \quad \ge 16. \quad =.$$
$$x_1, x_2 \ge 0$$

$$\boxed{\begin{aligned} x_1 + 2x_2 + x_3 &= 6 \\ 2x_1 + x_2 + x_4 &= 16 \end{aligned}}$$

$m = 2$
$n = 4$

$x_1, x_2, x_3, x_4 \ge 0.$

$n - m = 4 - 2 = 2$ fix to zero.

$$\binom{4}{2} = \frac{4!}{2!\,2!} = \frac{4.3}{2} = 6.$$

$$x_1 + 2x_2 + x_3 = 6$$
$$2x_1 + x_2 + x_4 = 16$$

① $x_1 = x_2 = 0 \Rightarrow x_3 = 6,\ x_4 = 16.$ feasible.

Soru yanlış verilmiş.

2. kısıt $\ge$ olacak.

*3. JoShop manufactures three products whose unit profits are \$2, \$5, and \$3, respectively. The company has budgeted 80 hours of labor time and 65 hours of machine time for the production of three products. The labor requirements per unit of products 1, 2, and 3 are 2, 1, and 2 hours, respectively. The corresponding machine-time requirements per unit are 1, 1, and 2 hours. JoShop regards the budgeted labor and machine hours as goals that may be exceeded, if necessary, but at the additional cost of \$15 per labor hour and \$10 per machine hour. Formulate the problem as an LP, and determine its optimum solution using TORA, Solver, or AMPL.

x_j = # of units of products 1, 2, and 3 **3**

Maximize $Z = 2x_1 + 5x_2 + 3x_3 - 15x_4^+ - 10x_5^+$

subject to

$$2x_1 + x_2 + 2x_3 + x_4^- - x_4^+ = 80$$

$$x_1 + x_2 + 2x_3 + x_5^- - x_5^+ = 65$$

all variables ≥ 0

Solution: $Z = \$325$

$x_2 = 65$ units, $x_4^- = 15$

All other variables $= 0$

continued...

$2x_1 + x_2 + 2x_3 \leq 80$ Labor +

$x_1 + x_2 + 2x_3 \leq 65$ Machine +

~~Slack ≥ 0.~~

$2x_1 + x_2 + 2x_3 + x_4 = 80$, x_4 is free.

free. $0 \Rightarrow 80$ ✓ No penalty.

$+ \Rightarrow < 80$ ✓ No penalty

$- \Rightarrow > 80$ penalty!

$x_4 = x_4^- - x_4^+$

$0 \quad + \Rightarrow$ penalty.

$2x_1 + x_2 + 2x_3 + x_4^- - x_4^+ = 80$; $x_4^-, x_4^+ \geq 0$

$x_1 + x_2 + 2x_3 + x_5 = 65.$, ~~$x_5 \geq 0$.~~

no penalty. $\left\{ \begin{array}{ll} \text{pos} & < 65 \\ 0 & = 65. \end{array} \right.$ x_5 is free.

No penalty. { 0 = 65.

penalty ← neg. > 65.

$$x_1 + x_2 + 2x_3 + x_5^- - x_5^+ = 65,$$

$$x_5^-, x_5^+ \geq 0.$$

Starting b.f.s = ?

1 0 $\quad (x_4^-, x_5^-)$

0 1 $\quad$ = 80, = 65

x_4^- $\quad x_5^-$

*1. For the following LP, identify three alternative optimal basic solutions, and then write a general expression for all the nonbasic alternative optima comprising these three basic solutions.

$$\text{Maximize } z = x_1 + 2x_2 + 3x_3$$

subject to

$$x_1 + 2x_2 + 3x_3 \leq 10$$
$$x_1 + x_2 \leq 5$$
$$x_1 \leq 1$$
$$x_1, x_2, x_3 \geq 0$$

Note: Although the problem has more than three alternative basic solution optima, you are only required to identify three of them. You may use TORA for convenience.

Point	Iteration	Basic	x_1	x_2	x_3	s_1	s_2	s_3	Solution	Ratio
	0	z	-1	-2	-3	0	0	0	0	
		s_1	1	2	$\boxed{3}$	1	0	0	10	→
		s_2	1	1	0	0	1	0	5	
		s_3	1	0	0	0	0	1	1	
	I	z	0	$\boxed{0}$	0	1	0	0	10	
$(0,0,\frac{10}{3})$		x_3	1/3	2/3	1	1/3	0	0	10/3	5 or
		s_2	1	$\boxed{1}$	0	0	1	0	5	5 →
		s_3	1	0	0	0	0	1	1	–
	II	z	$\boxed{0}$	0	0	1	0	0	10	
$(0,5,0)$		x_3	-1/3	0	1	1/3	-2/3	0	0	–
		x_2	1	1	0	0	1	0	5	5
		s_3	$\boxed{1}$	0	0	0	0	1	1	1 ⇒
	III	z	0	0	0	1	$\boxed{0}$	0	10	
$(1,4,\frac{1}{3})$		x_3	0	0	1	1/3	-2/3	1/3	1/3	–
		x_2	0	1	0	0	$\boxed{1}$	-1	4	⇒
		x_1	1	0	0	0	0	1	1	–
	IV	z	0	0	0	1	0	0	10	
$(1,0,3)$		x_3	0	2/3	1	1/3	0	-1/3	3	
		s_2	0	1	0	0	1	-1	4	⇒
		x_1	1	0	0	0	0	1	1	

Tableau I: z-row ≥ 0. optimal. Nonbasic $x_1, x_2 \Rightarrow 0$ z-row value.

Tableau II: alternative optimal. degenere

Tableau III: ⇒ alternative optimal.

Tableau IV: ⇒ alternative opt.

Three alternative basic optima:

$$(x_1, x_2, x_3) = \begin{cases} (0, 0, 10/3) & A_1 \\ (0, 5, 0) & A_2 \\ (1, 4, 1/3) & A_3 - A_4. \end{cases}$$

The associated <u>nonbasic</u> alternative optima are

$$\tilde{x}_1 = \lambda_3$$

$$\tilde{x}_2 = 5\lambda_2 + 4\lambda_3$$

$$\tilde{x}_3 = 10/3\,\lambda_1 + 1/3\,\lambda_3$$

where

$$\lambda_1 + \lambda_2 + \lambda_3 = 1$$

$$0 \le \lambda_i \le 1, \quad i = 1, 2, 3$$

$$\alpha x^1 + (1-\alpha) x^2.$$

$$\alpha, 1-\alpha \in [0,1]. \quad , \alpha + (1-\alpha) = 1.$$

$$\alpha x^1 + \beta x^2$$

$$\alpha, \beta \in [0,1] \quad , \alpha + \beta = 1.$$

$$x^1, x^2, x^3, \ldots, x^k. \to \text{alternative}$$

$$\alpha_1 x^1 + \alpha_2 x^2 + \alpha_3 x^3 + \ldots + \alpha_k x^k.$$

$$\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_k \in [0,1]$$

$$\sum^{k} \alpha_i = 1.$$

$$\sum_{i=1} \alpha_i = 1.$$

$$x = \lambda_1 x^1 + \lambda_2 x^2 + \lambda_3 x^3$$

$$= \lambda_1 \left(0, 0, \tfrac{10}{3}\right) + \lambda_2 (0, 5, 0)$$

$$+ \lambda_3 \left(1, 4, 1/3\right)$$

$$= \Big(\underbrace{\lambda_3}_{x_1} , \underbrace{5\lambda_2 + 4\lambda_3}_{x_2} , \underbrace{\tfrac{10}{3}\lambda_1 + \tfrac{1}{3}\lambda_3}_{x_3.} \Big)$$

$$\lambda_1, \lambda_2, \lambda_3 \in [0,1].$$

$$\sum \lambda_1 + \lambda_2 + \lambda_3 = 1.$$

***14.** Wyoming Electric Coop owns a steam-turbine power-generating plant. Because Wyoming is rich in coal deposits, the plant generates its steam from coal. This, however, may result in emission that does not meet the Environmental Protection Agency standards. EPA regulations limit sulfur dioxide discharge to 2000 parts per million per ton of coal burned and smoke discharge from the plant stacks to 20 lb per hour. The Coop receives two grades of pulverized coal, $C1$ and $C2$, for use in the steam plant. The two grades are usually mixed together before burning. For simplicity, it can be assumed that the amount of sulfur pollutant discharged (in parts per million) is a weighted average of the proportion of each grade used in the mixture. The following data are based on consumption of 1 ton per hour of each of the two coal grades.

Coal grade	Sulfur discharge in parts per million	Smoke discharge in lb per hour	Steam generated in lb per hour
$C1$ x_1	1800	2.1	12,000
$C2$ x_2	2100	.9	9,000

(a) Determine the optimal ratio for mixing the two coal grades.

(b) Determine the effect of relaxing the smoke discharge limit by 1 lb on the amount of generated steam per hour.

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x_1 = tons of C_1 consumed per hour

x_2 = tons of C_2 consumed per hour

Maximize $z = 12000x_1 + 9000x_2$

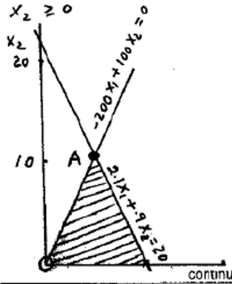
s.t.

$1800x_1 + 2100x_2 \le 2000(x_1 + x_2)$?

or $-200x_1 + 100x_2 \le 0$

$2.1x_1 + .9x_2 \le 20$

$x_1, x_2 \ge 0$



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